THE FRACTIONAL DE GIORGI CONJECTURE IN DIMENSION FIVE FOR SMALL ORDERS

Abstract. We prove one-dimensional symmetry for bounded entire solutions of $(-\Delta)^s U = U - U^3$ in $\mathbb{R}^5$ that are strictly increasing in one direction, for all sufficiently small $s > 0$. The geometric result is flatness of smooth stable nonlocal cones in $\mathbb{R}^4$ for small perimeter order $p = 2s$. At order zero, a signed estimate for the angular kernel and a spherical moment bound give a lower bound on the total conormal. A weighted stability test then forces the conormal to be constant. Cone flatness yields minimization of the monotone solution and flatness of its limiting interfaces. It also gives finite Morse index rigidity for the Allen–Cahn equation and a nonlocal Bernstein theorem in $\mathbb{R}^4$.

Draft status. This candidate proof was obtained by our team on November 10, 2026, through extensive dialogue-based training with AI. The team is currently working hard to verify all the proofs.

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1. Introduction and main results

The fractional De Giorgi problem asks when a bounded entire solution of the fractional Allen–Cahn equation is one-dimensional if it is strictly increasing in one direction. In that case its level sets are parallel hyperplanes. We use the Fourier normalization $\widehat{(-\Delta)^s u}(\xi) = |\xi|^{2s}\widehat{u}(\xi)$. The perimeter order is $p = 2s$. For $0 < s < 1/2$, Savin and Valdinoci showed that the rescaled phase-transition energies Γ-converge to a multiple of the nonlocal perimeter [12].

1.1. Main results.

We first state the symmetry result. It treats small s in dimension five with no hypothesis on the limits as $x_5 \to \pm\infty$.

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Theorem 1.1 (Monotone rigidity in dimension five)**.** *There exists $s_* \in (0,1/2)$ such that, for every $0 < s < s_*$, every bounded entire classical solution of*

$$(-\Delta)^s U = U - U^3 \quad \text{in } \mathbb{R}^5, \qquad \partial_{x_5} U > 0,$$

has the form

$$U(x) = q(e \cdot x - a), \qquad |e| = 1, \quad e_5 > 0, \quad a \in \mathbb{R}.$$

Here q is a strictly increasing classical solution of

$$(-\partial_t^2)^s q = q - q^3, \qquad q(0) = 0, \qquad \lim_{t \to \pm\infty} q(t) = \pm 1,$$

with the one-dimensional Fourier normalization.

The geometric part of the proof concerns cones one dimension lower. We call a cone nontrivial if both phases have positive measure. Its link is the intersection of its boundary with the unit sphere. The first variation of the localized p-perimeter vanishes at a stationary cone. At a stable cone its second variation is nonnegative. We test smooth deformations supported away from zero. The perimeter convention is given in (1).

Theorem 1.2 (Flat stable cones at small order)**.** *There exists $p_0 \in (0,1)$ such that, for $0 < p < p_0$, every nontrivial stationary stable p-perimeter cone in $\mathbb{R}^4$ with smooth compact link and locally one-sided boundary away from zero agrees almost everywhere with a halfspace.*

Together with Tsiamis's theorem in dimension three, the cone classification gives $n_p^* > 4$ for small p. Here n_p^* is the first dimension $n \geq 3$ admitting a nonflat smooth stable p-minimal cone in the punctured space, in the sense of Caselli–Florit-Simon–Serra [8, Definition 1.11]. Their finite Morse index theorems give the next two results. The number s_* in Theorem 1.1 is chosen so that both apply for $p = 2s < 2s_*$.

For the equation, the Morse index counts the largest dimension of a subspace of compactly supported smooth tests on which the energy second variation (25) is negative definite. The index of every solution in the next result is zero.

Corollary 1.3 (Finite Morse index rigidity in dimension four)**.** *Let $0 < s < s_*$. Every bounded entire classical solution of $(-\Delta)^s u = u - u^3$ in $\mathbb{R}^4$ with finite Morse index is a constant well $+1$ or -1, or has the form $u(x) = q(e \cdot x - a)$, where $e \in S^3$, $a \in \mathbb{R}$, and q is a normalized increasing layer as in Theorem 1.1. In each case u is stable.*

For a stationary phase, the geometric Morse index counts the largest dimension of a space of compactly supported smooth ambient vector fields on which the second variation of the p-perimeter is negative definite. Stability is the case of index zero. Thus the following result includes the stable nonlocal Bernstein theorem in dimension four.

Theorem 1.4 (Finite Morse index nonlocal Bernstein theorem in dimension four)**.** *Let $0 < p < 2s_*$ and let $E \subset \mathbb{R}^4$ be open with nonempty, locally one-sided C^2 boundary. If E is stationary for the p-perimeter and has finite geometric Morse index in $\mathbb{R}^4$, then E is an open halfspace and ∂E is an affine hyperplane.*

1.2. **Known results and current status.** In dimension two, Cabré and Sire proved symmetry for bounded stable solutions for every $0 < s < 1$ [6, Theorem 2.12]. In dimension three, Cabré and Cinti proved monotone symmetry for $1/2 \leq s < 1$ [3, Theorem 1.5]. Dipierro, Farina, and Valdinoci proved the result for $0 < s < 1/2$ [11, Theorem 1.1]. These three-dimensional results require no prescribed vertical limits.

For $s < 1/2$, Cabré, Cinti, and Serra related stable solutions to stable nonlocal cones [5, Theorem 1.8]. Their classification of smooth stable cones in $\mathbb{R}^3$ for p close to one [4] gives monotone rigidity in $\mathbb{R}^4$ near $s = 1/2$ from below [5, Corollary 1.11]. Chan, Dipierro, Serra, and Valdinoci proved flatness of smooth stable cones in $\mathbb{R}^4$ for p close to one [9, Theorem 1.5]. Together with the three-dimensional cone result and the reduction in Section 5, this gives

five-dimensional monotone rigidity near $s = 1/2$ from below. In this range, Caselli–Florit-Simon–Serra obtain the corresponding finite Morse index results in dimension four from the same cone classifications [8, Remark 1.33].

At the other end, Tsiamis proved flatness of smooth stable cones in $\mathbb{R}^3$ for small p [15, Theorem 1.1]. His compactness and pinching theorems apply in higher dimensions [15, Theorems 3.5 and 4.2]. We use them to pass from positive p to a balanced phase in S^3, and then back to positive p. The geometric step in Theorem 1.2 is to show that every such limiting phase in S^3 is a hemisphere.

1.3. **The proof and the order of the sections.** The conormal of a hemispherical phase is constant. We measure its alignment on a general link by the length of the integral of the conormal divided by the link area. This number lies between zero and one. The constant stability test gives a signed row estimate. A bound on the first two spherical moments then excludes small values of the normalized integral. For the other values, we test stability with the difference between one and the conormal component in the mean direction. The cap estimate keeps the normal defect inside the integral. This allows us to absorb the term that comes from nearby pairs and force equality in the triangle inequality for the conormal.

Section 2 gives the two stability tests at order zero. Section 3 gives the moment and row estimates. Section 4 combines them with the cap bound and proves Theorem 1.2. Section 5 first derives the finite Morse index results from the cone theorem. We then classify the stable vertical limits of U in dimension five. This gives minimization of U. The limiting interfaces are halfspaces, and improvement of flatness gives Theorem 1.1.

2. Stable cones and their spherical limits

We count each pair of points in opposite phases once. For an open bounded set $D \subset \mathbb{R}^n$, the perimeter of a measurable phase E in D is

$$\mathrm{Per}_p(E; D) = \iint_{(E\cap D)\times E^c} \frac{dx\,dy}{|x-y|^{n+p}} + \iint_{(E\setminus D)\times (E^c\cap D)} \frac{dx\,dy}{|x-y|^{n+p}}. \tag{1}$$

A cone is invariant under all positive dilations about zero. We use the representative made up of points of density one in the phase. In a one-sided boundary chart this is the open side of the chart. At the vertex of a nontrivial cone the density is strictly between zero and one. Smooth changes of variables preserve null sets, so this choice does not affect the perimeter or its variations.

Lemma 2.1 (Localized perimeter and its variations)**.** *Let $n \geq 2$ and $0 < p < 1$. Suppose $E \subset \mathbb{R}^n$ is open with locally one-sided C^2 boundary, or is a cone with a smooth compact link and locally one-sided boundary off zero. The localized p-perimeter is finite on bounded open sets and is twice differentiable along compactly supported smooth deformations, supported away from zero in the cone case. At stationarity the second derivative is a quadratic form of the initial velocity. Stationarity implies zero pointwise nonlocal mean curvature on the regular boundary. For a cone, stability also gives the normal Jacobi inequality of [15, Section 2].*

Proof. In the C^2 case, finitely many graph charts cover the portion of the boundary in each closed ball. Each chart has locally finite area, and the one-sided condition allows Gauss–Green in the phase. This gives finite classical perimeter on bounded balls. On a cone the classical area element is $r^{n-2}\,dr\,d\mu$, where μ is link area. This gives finite classical perimeter on bounded balls. In applying Gauss–Green across the origin, the flux through an inner sphere of radius r is $O(r^{n-1})$. Thus there is no additional perimeter at zero. For small z, the L^1 difference of $\mathbf{1}_E(\cdot + z)$ and $\mathbf{1}_E$ on a ball is at most $|z|$ times the classical perimeter on a larger ball. Integration against $|z|^{-n-p}$ gives the finite radial integral $\int_0^1 r^{-p}\,dr$. At large distances we use $\int_1^\infty r^{-1-p}\,dr < \infty$.

Choose D to contain the support of the deformation in its interior. Pull the deformed perimeter back to the original pairs. The flow is uniformly bi-Lipschitz for small time. Its

velocities and accelerations have Lipschitz bounds in the initial position. For $K(z) = |z|^{-n-p}$, we have $|D^j K(z)| \le C|z|^{-n-p-j}$ for $j = 1, 2$. The chain rule pairs DK with one velocity or acceleration difference, and D^2K with two velocity differences. Each difference is $O(|x-y|)$. The Jacobians and their first two derivatives are bounded as well. Thus each differentiated pair integrand is bounded by $C|x-y|^{-n-p}$ on opposite-phase pairs in the localization. This justifies differentiation under the integral. The derivatives are independent of D, because the extra pairs in a larger domain have both endpoints outside D. The same argument works for a bounded Lipschitz D when an autonomous field has support in $\overline{D}$. Such a field vanishes on D^c by continuity from the exterior, so its flow fixes D^c. One can differentiate in a larger ball and use the constant contribution from the extra pairs.

For two deformations with the same initial velocity, the terms quadratic in that velocity agree in their second derivatives. The remaining terms are linear in the initial acceleration, in the same way that the first derivative is linear in the initial velocity. Hence the difference of the second derivatives is the first variation evaluated on the difference of the initial accelerations. It vanishes at stationarity. Taking the deformation $\Phi_t(x) = x + tX(x)$ for a compact smooth field X shows the quadratic dependence on X. This is a diffeomorphism for $|t|\|DX\|_\infty < 1$, by solving for the inverse by contraction. Its acceleration is zero, and the kernel and Jacobian terms of order two are quadratic in X and its first derivatives.

Use the curvature

$$H_p[E](x) = \frac{1}{2}\,\mathrm{PV}\int_{\mathbb{R}^n} \frac{\mathbf{1}_{E^c}(y) - \mathbf{1}_E(y)}{|x-y|^{n+p}}\,dy.$$

On a compact regular patch, subtract the signed indicator of the tangent halfspace at x. Its integral on each centered annulus vanishes by reflection. The remaining difference has volume $O(r^{n+1})$ in a ball of radius r, by the quadratic graph bound. Summation over dyadic annuli gives a uniform $O(r^{1-p})$ bound for the remaining near part. The same bound holds for the regularized kernel $(|x-y|^2+\eta^2)^{-(n+p)/2}$. Its curvature converges locally uniformly to $H_p[E]$.

For the regularized kernel the first variation is $2\int H_{p,\eta}[E]\, X\cdot\nu\, d\sigma$, where X is the initial velocity. This follows by writing the two pullback derivatives as divergences and applying Gauss–Green in each phase. The signs are opposite in the two phases. The bounds $|D^j K_\eta(z)| \le C|z|^{-n-p-j}$ for $0 \le j \le 2$ hold uniformly in η. The pair majorant therefore permits $\eta \downarrow 0$ in both perimeter derivatives. Stationarity gives $\int H_p[E]\, X\cdot\nu\, d\sigma = 0$ for each smooth field supported on the patch. At a regular point x_0, the scalar product $\nu(x_0)\cdot\nu(x)$ is positive near x_0. If the continuous curvature $H_p[E]$ were nonzero at x_0, a nonnegative bump times the constant field $\nu(x_0)$, supported in a neighborhood where the curvature has one sign, would give a nonzero first variation. Thus $H_p[E] = 0$ there.

For the cone Jacobi form, take a compactly supported normal speed h on $M = \partial E \setminus \{0\}$. Extend $h\nu$ to a smooth ambient field by tubular projection and a cutoff. Write $K_\eta(z) = (|z|^2+\eta^2)^{-(n+p)/2}$. In differentiating the regularized curvature at a moving boundary point X, the moving phase contributes $-\int_M h(Y)K_\eta(X-Y)\,d\sigma_Y$. The motion of X contributes $\int_M h(X)(\nu_X\cdot\nu_Y)K_\eta(X-Y)\,d\sigma_Y$, by $\nabla_X K_\eta(X-Y) = -\nabla_Y K_\eta(X-Y)$ and Gauss–Green. The terms from differentiating the surface element and the speed in the first variation tend to zero, since the regularized curvature converges uniformly to zero on the support. Multiplying the two curvature terms by $h(X)$ and symmetrizing shows that half the second variation is the limit of

$$\frac{1}{2}\iint_{M\times M}(h(X)-h(Y))^2K_\eta(X-Y)\,d\sigma_X d\sigma_Y - \iint_{M\times M} h(X)^2(1-\nu_X\cdot\nu_Y)K_\eta(X-Y)\,d\sigma_X d\sigma_Y.$$

For pairs near the diagonal with a nonzero term, both points lie in a compact regular region and the numerators are $O(|X-Y|^2)$. At infinity one point lies in the support of h, and the integral in the other is bounded using the area element $r^{n-2}\,dr\,d\mu$. We can therefore replace K_η by K. This is the form in [15, Section 2], and ambient stability makes it nonnegative. □

2.1. **The data at order zero.** The spherical phase of a cone in $\mathbb{R}^4$ lies in S^3. For a finite-perimeter phase $\Omega \subset S^3$, write ω for spherical volume, $\mu = |D_{S^3}\mathbf{1}_\Omega|$, and ν for the outward unit conormal. Thus $D_{S^3}\mathbf{1}_\Omega = -\nu\mu$. Put

$$L = \mu(S^3), \qquad V = \int \nu \, d\mu, \qquad M = \int x \otimes \nu_x \, d\mu_x. \tag{2}$$

When $L > 0$, let $v = |V|/L$. The triangle inequality gives $0 \le v \le 1$. A hemisphere has $v = 1$, as shown in Figure 1.

For $-1 \le z < 1$ and $\rho > 0$, set $D_z(\rho) = 1 + \rho^2 - 2\rho z$. We use the kernels

$$A(z) = \int_0^\infty \frac{\rho \, d\rho}{D_z(\rho)^2}, \qquad B(z) = \int_0^\infty \frac{\rho^2 \, d\rho}{D_z(\rho)^2}, \qquad W(z) = B(z) - A(z). \tag{3}$$

These are the angular kernels at order zero. Define the normal defect and its integral by

$$P_x = \int B(x \cdot y)(1 - \nu_x \cdot \nu_y) \, d\mu_y, \qquad P = \int P_x \, d\mu_x. \tag{4}$$

All angular double integrals use $\mu \otimes \mu$ and omit the diagonal. We write $B_r(x)$ for a spherical cap defined by chord distance when the center and the domain of integration lie in S^3.

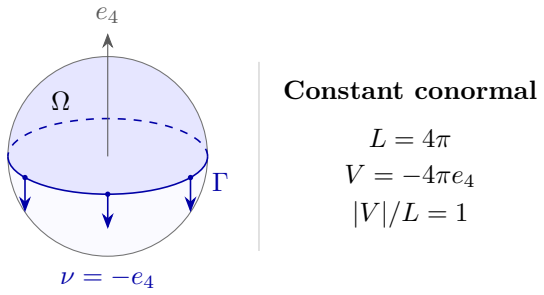


FIGURE 1. The halfspace link. For $E = \{x_4 > 0\}$, the spherical phase $\Omega = E \cap S^3$ is a hemisphere and its boundary Γ is a great two-sphere. The arrows show the outward conormal. The sphere in the drawing is a lower-dimensional model of S^3.

Proposition 2.2 (Limit tests)**.** *Let $p_i \downarrow 0$, with $0 < p_i < 1$, and let $E_i \subset \mathbb{R}^4$ be nontrivial cones, stationary and stable at order p_i, with smooth compact links and locally one-sided boundary off zero. Along a subsequence the spherical phases converge strictly in BV to a phase Ω of volume π^2. This means that their indicators converge in $L^1(S^3)$ and their perimeters converge. The link measures μ_i and the limit μ obey the uniform cap bound $\mu_i(B_r(x)), \mu(B_r(x)) \le Cr^2$ for $x \in S^3$ and $0 < r \le 2$. The measures $(x, \nu_{i,x})_\# \mu_i$ converge weakly to $(x, \nu_x)_\# \mu$. There is a finite number J with*

$$P \le J \le \iint W(x \cdot y) g_a(x) g_a(y) \, d\mu_x d\mu_y, \qquad g_a(x) = 1 - a \cdot \nu_x \tag{5}$$

for every $a \in \mathbb{R}^4$. For each unit vector e one also has

$$\int g_e(x) P_x \, d\mu_x \le J. \tag{6}$$

Proof. Fix $\bar{p} \in [p_1, 1)$. The links are closed embedded hypersurfaces of S^3. In fact the radial direction is tangent to the regular cone boundary and normal to S^3, so the intersection is transverse. A nontrivial cone has a nonempty link since the punctured space is connected. Lemma 2.1 gives the stationarity and stability hypotheses in [15, Theorem 3.5]. That theorem gives strict BV convergence and half volume. Its cap bound, equation (25) there, passes to the limit on open caps. The measures $(x, \nu_{i,x})_\# \mu_i$ on $S^3 \times S^3$ converge weakly to $(x, \nu_x)_\# \mu$.

The same theorem gives weak convergence of $P_{i,x}\,d\mu_i(x)$ to a finite measure ℓ. Here $P_{i,x}$ uses the order p_i kernel

$$B_{p_i}(z) = \int_0^\infty \frac{\rho^2\,d\rho}{D_z(\rho)^{(4+p_i)/2}}$$

and the conormals of the i-th link in (4). With $J = \ell(S^3)$, Corollary 3.6 of [15] gives (5).

For (6), use the translation identity $\int (e\cdot\nu_i)P_{i,x}\,d\mu_i = 0$ from [15, Lemma 2.4, equation (8)]. Thus $\int(1 - e\cdot\nu_i)P_{i,x}\,d\mu_i \to J$. Write $\eta_i = (x,\nu_{i,x})_\#\mu_i$ and $\eta = (x,\nu_x)_\#\mu$. For a positive integer k, use $\chi_k(x,y) = \min\{1, \max\{0, k|x-y| - 1\}\}$. The integrand is nonnegative, so

$$\iint \chi_k(x,y)B_{p_i}(x\cdot y)(1-\xi\cdot\zeta)(1-e\cdot\xi)\,d\eta_i(x,\xi)d\eta_i(y,\zeta) \le \int (1-e\cdot\nu_i)P_{i,x}\,d\mu_i.$$

Extend the cutoff integrand by zero on the diagonal, where the cutoff vanishes in a neighborhood. On its support $B_{p_i} \to B$ uniformly. In the radial formula the small-ρ tail is bounded by $C\rho^2$ and the large-ρ tail by $C\rho^{-2}$. On the middle radial interval the denominator stays away from zero off the angular diagonal. This proves the uniform convergence. The products $\eta_i \otimes \eta_i$ converge weakly to $\eta\otimes\eta$, since finite sums of products of continuous functions are dense among continuous functions on the product. The cutoff integral for the limit is therefore at most J. The cap bound gives $\mu(\{x\}) = 0$ for every x, so the product measure assigns no mass to the diagonal. Monotone convergence as $k\to\infty$ proves (6). □

3. Area, moments, and signed rows

We now work with the data in (2)–(4). The area estimate below retains two moments of the phase. The second moment will control a negative term from the kernel estimate.

Lemma 3.1 (Two moments of the spherical phase)**.** *Suppose $|\Omega| = \pi^2$. Then*

$$L \ge a - b|V|^2 + c|M|_F^2, \qquad a = \frac{25\pi}{6}, \quad b = \frac{5(3\sqrt{5}-5)}{27\pi^3}, \quad c = \frac{5\sqrt{5}}{8\pi^3}. \tag{7}$$

Here $|\cdot|_F$ is the Frobenius norm. The matrix M is symmetric and trace free. In particular,

$$\iint (x\cdot y)(1-\nu_x\cdot\nu_y)\,d\mu_x d\mu_y = \left|\int x\,d\mu_x\right|^2 - |M|_F^2. \tag{8}$$

Proof. For $\theta = 2\pi/5$ let T_θ be averaging over points at spherical distance θ. If ξ is a uniform unit vector in $x^\perp$, the other endpoint is $y = \cos\theta\,x + \sin\theta\,\xi$. For a smooth f, integrate its directional derivative along this geodesic. The uniform law of the ordered pair (x,ξ) is invariant under geodesic flow. In a three-dimensional tangent space the mean absolute scalar product of a unit vector with a uniform unit direction is $1/2$. It follows that the average difference $|f(x) - f(y)|$, integrated in x, is at most $(\theta/2)\int|\nabla_{S^3}f|$. Apply smooth BV approximation to $\mathbf{1}_\Omega$. Both endpoint marginals are uniform, so the error in the average difference is bounded by twice the L^1 approximation error.

Let $u = \mathbf{1}_\Omega - 1/2$, and denote its degree k spherical harmonic projection by u_k. Its mean is zero and $\|u\|_2^2 = \pi^2/2$. The square of an indicator difference is its absolute value. Expanding the square and using the two uniform marginals gives $2\langle u, (I - T_\theta)u\rangle$ for the integrated square difference. The geodesic bound therefore gives

$$\frac{\theta L}{4} \ge \langle u, (I - T_\theta)u\rangle.$$

The eigenvalue of T_θ in degree k is $\lambda_k = \sin((k+1)\theta)/((k+1)\sin\theta)$. In geodesic polar coordinates about x, the spherical Laplacian is $\partial_\theta^2 + 2\cot\theta\,\partial_\theta + (\sin\theta)^{-2}\Delta_{S^2}$. The angular Laplacian integrates to zero. Thus the mean F of a degree k harmonic Y solves $F'' + 2\cot\theta\,F' + k(k+2)F = 0$. The function $\sin\theta\,F(\theta)$ then solves $G'' + (k+1)^2G = 0$ with $G(0) = 0$ and $G'(0) = Y(x)$. This gives the eigenvalue formula. In particular, $\lambda_1 = \cos\theta$ and $\lambda_2 = -1/(6\cos(\pi/5))$. In degrees $3, 4, 5$ the values are negative, zero, and $1/6$. For $k \ge 6$, their absolute values are bounded by $1/(7\sin\theta) < 1/6$. Here $\sin\theta > \sin(\pi/3) > 6/7$.

Gauss–Green with the tangential part of a constant vector, whose divergence is -3 times its radial component, gives

$$V = -3\int_\Omega x\,d\omega. \tag{9}$$

Applying it to $x_i\nabla_{S^3}x_j$ gives

$$M_{ij} = \int_\Omega (\delta_{ij} - 4x_ix_j)\,d\omega = -4\int_{S^3} u\,x_ix_j\,d\omega.$$

This proves the symmetry and trace assertion. The coordinate functions are orthogonal with squared norm $\pi^2/2$, so (9) gives $u_1(x) = -2V\cdot x/(3\pi^2)$ and $\|u_1\|_2^2 = 2|V|^2/(9\pi^2)$. For a symmetric trace-free Q,

$$\int_{S^3}(x\cdot Qx)^2\,d\omega = \frac{\pi^2}{6}|Q|_F^2.$$

For this identity, rotation by $\pi/4$ in a coordinate plane gives $\int x_1^4 = 3\int x_1^2x_2^2$. Integrating $(\sum_i x_i^2)^2 = 1$ fixes these constants. Polarization and the formula for M now give $u_2(x) = -3x\cdot Mx/(2\pi^2)$ and $\|u_2\|_2^2 = 3|M|_F^2/(8\pi^2)$.

Bound λ_k by $\sigma = 1/6$ for $k\ge 3$. The geodesic inequality yields

$$\frac{\theta L}{4} \ge (1-\sigma)\frac{\pi^2}{2} - (\cos\theta-\sigma)\frac{2|V|^2}{9\pi^2} + (\sigma-\lambda_2)\frac{3|M|_F^2}{8\pi^2}.$$

Substitution of $\cos(2\pi/5) = (\sqrt5-1)/4$ and $\cos(\pi/5) = (\sqrt5+1)/4$ gives (7). Finally, expanding the scalar products in the double integral gives (8). □

Lemma 3.2 (The kernels at order zero)**.** *For the kernels in* (3)*,*

$$\begin{aligned} W(z) &= \int_0^1 \frac{(1-\rho)^2}{D_z(\rho)^2}\,d\rho, \qquad \int_{-1}^1 W(z)\,dz = 1,\\ \int_{-1}^0 W(z)\,dz &= 1-\frac{\pi}{4}, \qquad W(0) = \frac{\pi}{4}-\frac12. \end{aligned} \tag{10}$$

For $-1<z<1$, let $T(z) = (\pi-\arccos z)/\sqrt{1-z^2}$. Then

$$A = \tfrac12 T', \qquad B = \tfrac12(T+zT'), \qquad A-zB = \tfrac12, \qquad W = \frac{T-1}{2(1+z)}. \tag{11}$$

For all $-1\le z<1$,

$$A(z)\ge u_0+q_0z, \qquad B(z)\ge \tfrac23+q_1z, \qquad B(z)\ge\tfrac13, \tag{12}$$

where $u_0 = 2\pi/(9\sqrt3)$, $q_0 = 8\pi/(9\sqrt3)-4/3$, and $q_1 = 2-8\pi/(9\sqrt3)$.

Proof. Change ρ to $1/\rho$ above $\rho=1$. On $(0,1)$ the combined numerators of A and B are 2ρ and $1+\rho^2$. Their difference gives the formula for W. Integrating it in z over $(-1,1)$ and $(-1,0)$ gives the respective radial integrands $2/(1+\rho)^2$ and $2/(1+\rho)^2-1/(1+\rho^2)$. At $z=0$, write $(1-\rho)^2 = 1+\rho^2-2\rho$. These computations yield (10).

Completing the square in $\int_0^\infty D_z^{-1}\,d\rho$ gives T. Its derivative is $2A$. Radial inversion gives $\int_0^\infty D_z^{-2}\,d\rho = B$, so $T = 2B-2zA$. Integrate $\partial_\rho(D_z^{-1})$ to get $A-zB = 1/2$. This proves (11). The functions A, B are convex by differentiation of the radial integrals. Using $(1-z^2)T' = 1+zT$ in (11), their slopes at $-1/2$ are q_0, q_1, and their values there are $u_0-q_0/2$, $2/3-q_1/2$, respectively. The slopes are positive by differentiation of the radial integrals. This gives the two tangent bounds. Since B is increasing, its minimum is $B(-1) = \int_0^\infty \rho^2(1+\rho)^{-4}\,d\rho = 1/3$. All endpoint formulas used here extend at -1 by continuity. □

Lemma 3.3 (A signed row estimate)**.** *Suppose that* $\mu(B_r(x)) \le Cr^2$ *for* $0 < r \le 2$ *and* $x \in S^3$*. Then*

$$\iint W(x\cdot y)\nu_x\cdot\nu_y\,d\mu_x d\mu_y \le (\pi^2-2\pi)L + W(0)|V|^2. \tag{13}$$

Proof. Fix $x \in S^3$ and a unit vector $a \perp x$. For $z = x\cdot y$ let $D(z) = 3W(z) + zW'(z)$ and $a^T(y) = a - (a\cdot y)y$. We have $\nabla_{S^3} z\cdot a^T = -z(a\cdot y)$ and $\operatorname{div}_{S^3} a^T = -3(a\cdot y)$. Hence the spherical divergence of $W(z)a^T(y)$ equals $-D(z)(a\cdot y)$. For $r = |x-y|$, the folded formula gives $W = O(r^{-1})$ and $W' = O(r^{-3})$. To see this, use $D_z(\rho) = (1-\rho)^2 + \rho r^2$ and rescale $1-\rho$ by r on $[1/2, 1]$. The other radial part is bounded. Since $|a\cdot y| \le r$, the divergence is integrable in spherical volume. Cutting the field off between $r = \varepsilon$ and $r = 2\varepsilon$ adds a term of size $O(\varepsilon^{-2})$ on a region of volume $O(\varepsilon^3)$. The cap bound gives $\int_{B_{2\varepsilon}(x)} W(x\cdot y)\,d\mu_y = O(\varepsilon)$, by summing $O(r^{-1})\mu(B_r(x))$ over dyadic radii. In particular the rows of W have uniformly bounded integrals. Gauss–Green yields

$$\int W(x\cdot y)a\cdot\nu_y\,d\mu_y = -\int_\Omega D(x\cdot y)(a\cdot y)\,d\omega_y. \tag{14}$$

The function D is positive and increasing. On $z \ge 0$ this follows from the positive radial derivatives. For $z < 0$, the folded formula and $D_z(\rho) \ge 2\rho(1-z)$ give $W' \le 2W/(1-z)$ and $W'' \le 3W'/(1-z)$. Since $z/(1-z) \ge -1/2$, both D and $D' = 4W' + zW''$ are positive there.

Compare (14) with the same identity for $H = \{y \in S^3 : a\cdot y < 0\}$. Its equator contains x and has constant conormal a. Its row is $2\pi\int_{-1}^1 W(z)\,dz = 2\pi$. The two parts of the mismatch are shown in Figure 2. The row for Ω is this value minus

$$\int_{\Omega\triangle H} D(x\cdot y)|a\cdot y|\,d\omega_y.$$

By (9), the weighted volume of the mismatch is

$$\int_{\Omega\triangle H}|a\cdot y|\,d\omega_y = \int_H(-a\cdot y)\,d\omega_y + \int_\Omega a\cdot y\,d\omega_y = \frac{4\pi - a\cdot V}{3}.$$

The last equality uses (9) for both phases Ω and H, since H has flux $4\pi a$. In the variable $z = x\cdot y$, the measure $|a\cdot y|\,d\omega_y$ has density $2\pi(1-z^2)\,dz$. This follows by writing $y = zx + \sqrt{1-z^2}\,\xi$ and using $\int_{S^2\subset x^\perp}|a\cdot\xi|\,d\xi = 2\pi$. The part $z < 0$ has weighted volume $4\pi/3$.

The function $D - D(0)$ is nonpositive on $z < 0$ and nonnegative on $z > 0$. Thus its integral over the mismatch is at least its integral over $z < 0$. Integration by parts gives

$$2\pi\int_{-1}^0 D(z)(1-z^2)\,dz = 4\pi\int_{-1}^0 W(z)\,dz = 4\pi - \pi^2.$$

The boundary term $z(1-z^2)W(z)$ vanishes at both endpoints. As $D(0) = 3W(0)$, the mismatch integral is at least $4\pi - \pi^2 - W(0)a\cdot V$. Set $a = \nu_x$ and integrate the resulting row bound in x. The uniform row bound just proved allows Fubini, yielding (13). □

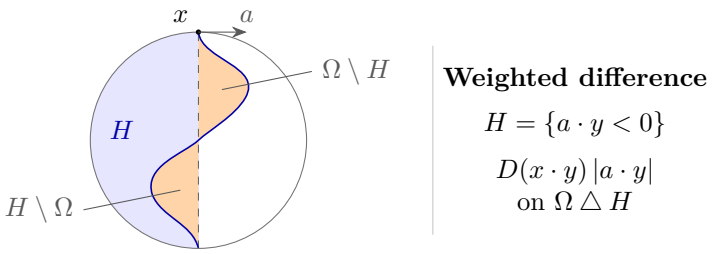


FIGURE 2. The signed row compares the phase Ω with a hemisphere H whose boundary passes through x. The colored regions show where the phases differ. Each part of the difference contributes a nonnegative amount to the deficit in the row for Ω. The disk is a schematic view in the coordinates $(a \cdot y, x \cdot y)$.

4. The weighted test and the cone theorem

We now show that a limit phase from Proposition 2.2 has constant conormal. The test with $a = 0$ gives a lower bound on the normalized flux v. We then use the test in the direction of V to reach the equality case $v = 1$.

4.1. The estimate on a cap.

Lemma 4.1 (Weighted cap bound)**.** *Assume the cap bound in Proposition 2.2 and $0 < v < 1$. Take $e = V/|V|$, $g = 1 - e \cdot \nu$, and $m = L(1 - v)$. If $P \le J$, $\int g P_x \, d\mu_x \le J$, and $J \le \iint W g(x) g(y) \, d\mu_x d\mu_y$, then*

$$\frac{23}{27}\frac{P}{m} \le w_1 m + C_1, \qquad w_1 = \frac{4\pi}{9\sqrt{3}} - \frac{1}{3}, \qquad C_1 = \frac{4\pi^2}{9\sqrt{3}} + \frac{2\pi}{3}. \tag{15}$$

Proof. Let $d\lambda = g \, d\mu$, so $\lambda(S^3) = m$ and $0 \le g \le 2$. For μ-almost every x and $0 < r \le 1$, we first prove

$$\lambda(B_r(x)) \le 2\pi r^2 + 2 \int_{B_r(x)} (1 - \nu_x \cdot \nu_y) \, d\mu_y. \tag{16}$$

Put $a = \nu_x$ and $\theta = 2 \arcsin(r/2)$. Write $y = \cos t \, x + \sin t \, \xi$ with $\xi \in S^2 \subset x^\perp$. Test Gauss–Green with the tangential projection of a times a smooth decreasing cutoff $\eta_\varepsilon(t)$. Take it equal to one for $t \le \theta - \varepsilon$ and zero for $t \ge \theta$. The divergence is

$$-3\eta_\varepsilon(t) \sin t \, (a \cdot \xi) + \eta_\varepsilon'(t) \cos t \, (a \cdot \xi).$$

Here the radial derivative of $a \cdot y = \sin t \, (a \cdot \xi)$ is $\cos t \, (a \cdot \xi)$. The volume element is $\sin^2 t \, dt \, d\xi$, and each sign part of $a \cdot \xi$ has integral π over S^2. Bound both divergence terms by their positive parts and let $\varepsilon \downarrow 0$. Since $\cos\theta \ge 0$, this gives

$$a \cdot \int_{B_r(x)} \nu_y \, d\mu_y \le \pi \left(3 \int_0^\theta \sin^3 t \, dt + \cos\theta \sin^2\theta \right) = 2\pi(1 - \cos\theta) = \pi r^2.$$

The measure $-\eta_\varepsilon'(t) \, dt$ has mass one and is supported in $[\theta - \varepsilon, \theta]$. The boundary integral converges to the integral on the open cap by bounded convergence, even if its edge has positive μ-measure. Add the integral of $1 - \nu_x \cdot \nu_y$ to the flux estimate and use $g \le 2$. This proves (16).

Write $w(r) = W(1 - r^2/2)$. In the folded formula for W put $u = (1 - \rho)/\sqrt{\rho}$. Then

$$w(r) = \int_0^\infty \frac{2u^2}{\sqrt{u^2 + 4}(u^2 + r^2)^2} \, du, \qquad -w'(r) \le 4r \int_0^\infty \frac{u^2}{(u^2 + r^2)^3} \, du = \frac{\pi}{4r^2}.$$

The inequality uses $2/\sqrt{u^2 + 4} \le 1$. The last integral is evaluated by $u = r \tan t$. From (11),

$$B(\cos\vartheta) = \frac{\pi - \vartheta + \cos\vartheta \sin\vartheta}{2 \sin^3\vartheta} \qquad (0 < \vartheta < \pi).$$

For $r = 2\sin t$ the bound $r^3 B(1-r^2/2) \geq \pi/2$ is equivalent to $\pi - 2t + \cos(2t)\sin(2t) \geq \pi\cos^3 t$. The difference is zero at $t = 0, \pi/2$. Its derivative is $\sin t \cos^2 t(3\pi - 16\sin t)$, so it increases and then decreases. The bound extends at the antipode by continuity. For $0 < r < 1$ it follows that

$$\frac{w(r) - w(1)}{B(1 - r^2/2)} \leq \tfrac{1}{2} r^2(1 - r) \leq \frac{2}{27}. \tag{17}$$

Since w is decreasing, $w(|x-y|) \leq w(1) + \int_0^1 (-w'(r))\mathbf{1}_{\{|x-y|<r\}}\, dr$. Integration in y gives

$$\int w(|x-y|)\, d\lambda_y \leq w(1)m + \int_0^1 (-w'(r))\lambda(B_r(x))\, dr.$$

Apply (16) and integrate in x against λ. By Tonelli the normal defect term is

$$2 \iint_{|x-y|<1} [w(|x-y|) - w(1)](1 - \nu_x \cdot \nu_y) g(x)\, d\mu_x d\mu_y.$$

By (17) it is at most $(4/27)\int g P_x\, d\mu_x \leq 4J/27$. The weighted test and $P \leq J$ give

$$\frac{23}{27} P \leq \frac{23}{27} J \leq w(1)m^2 + 2\pi m \int_0^1 r^2(-w'(r))\, dr.$$

Here $w(1) = W(1/2) = w_1$ by (11). Also

$$\int_0^1 r w(r)\, dr = \int_{1/2}^1 W(z)\, dz = \int_0^1 \frac{d\rho}{2(1 + \rho^2 - \rho)} = \frac{\pi}{3\sqrt{3}}.$$

The middle equality uses the folded formula and the last uses completion of the square. The boundary term $r^2 w(r)$ tends to zero at zero. Integration by parts now gives $2\pi \int_0^1 r^2(-w') = C_1$. Division by $m > 0$ proves the claim. □

4.2. **The flux inequalities.** The following rational intervals will be used in the comparisons.

$$\frac{333}{106} < \pi < \frac{355}{113}, \qquad \frac{265}{153} < \sqrt{3} < \frac{1351}{780}, \qquad \frac{559}{250} < \sqrt{5} < \frac{2237}{1000}. \tag{18}$$

The square-root bounds follow by squaring. For the bounds on π, use $\pi = 16\arctan(1/5) - 4\arctan(1/239)$. The tangent addition formula verifies the identity, with the angle after division by four in $(0, \pi/2)$. The four- and five-term alternating sums at $1/5$ give lower and upper bounds. At $1/239$ use the two- and one-term sums. The stated rational bounds follow by cross multiplication. All terminating decimals below denote exact rational numbers.

Proposition 4.2 (Rigidity of the limit phase)**.** *A half-volume phase with the cap bound and the inequalities of Proposition 2.2 is a hemisphere up to null sets.*

Proof. The spherical isoperimetric inequality gives $L \geq 4\pi$ for a half-volume phase [2, equation (2.2)]. Set $j = |M|_F^2/L^2$.

The constant test. Use (5) with $a = 0$. Since $W = B - A$, this gives

$$\iint A(x \cdot y)(1 - \nu_x \cdot \nu_y)\, d\mu_x d\mu_y \leq \iint W(x \cdot y)\nu_x \cdot \nu_y\, d\mu_x d\mu_y.$$

Apply the tangent estimate for A, the positional identity (8), and the signed row bound (13). With $d_0 = 4\pi - \pi^2$ and $w_0 = W(0)$, the result is

$$2\pi \geq d_0 + u_0 L(1 - v^2) - q_0 L j - w_0 L v^2.$$

Here $q_0 > 0$. Eliminate j using (7). We obtain

$$2\pi \geq d_0 - \frac{q_0}{c} + \left[u_0(1 - v^2) - \left(w_0 + \frac{q_0 b}{c}\right) v^2\right] L + \frac{q_0 a}{cL}. \tag{19}$$

From (18) and the exact constants in (7) and (12),

$$d_0 > 2.69, \qquad 6.15 < q_0/c < 6.20, \qquad a > 13.08,$$
$$u_0 > 0.403, \qquad b < 0.01022, \qquad w_0 < 0.286.$$

If $v \le 9/25$, the coefficient of L in (19) is greater than

$$0.403\big(1-(9/25)^2\big) - (0.286 + 6.20 \cdot 0.01022)(9/25)^2 > 0.305.$$

Also $q_0 a/c > 80$. The arithmetic–geometric mean inequality bounds the last two terms in (19) below by $2\sqrt{0.305 \cdot 80} > 197/20$. Thus the right-hand side is greater than $2.69 - 6.20 + 197/20 = 6.34 > 2\pi$. We conclude that $v > 9/25$.

The weighted test. If $v = 1$, put $e = V/L$. Then $|e| = 1$ and $\int |\nu - e|^2 \, d\mu = 2L - 2e \cdot V = 0$, so the conormal equals e almost everywhere. Tangency to the sphere gives $x \cdot e = 0$ for μ-almost every x. The indicator has zero derivative on each open hemisphere cut out by this equator. It is constant almost everywhere on each connected side, and half volume forces different constants. Thus it suffices to exclude $9/25 < v < 1$.

Put $f_0 = 23/27$ and apply Lemma 4.1 with $e = V/|V|$. Its hypotheses follow from (5) and (6). We get

$$f_0 \frac{P}{L(1-v)} \le w_1 L(1-v) + C_1. \tag{20}$$

We use three ranges of v.

First range, $9/25 \le v \le 13/20$. The tangent estimate for B and (8) give $P \ge L^2[\frac{2}{3}(1 - v^2) - q_1 j]$. Insert this in (20) and eliminate j with (7). The result is

$$C_1 \ge -D + kL + \frac{Da}{L}, \qquad \alpha = \frac{f_0 q_1}{c}, \quad D = \frac{\alpha}{1-v}, \quad k = \frac{2f_0}{3}(1+v) - w_1(1-v) - \frac{\alpha b v^2}{1-v}. \tag{21}$$

Consider the quadratic

$$F(v) = 4a\alpha \left[(1-v)\left(\frac{2f_0}{3}(1+v) - w_1(1-v) \right) - \alpha b v^2 \right] - [C_1(1-v) + \alpha]^2.$$

The bracket has negative quadratic coefficient. Subtracting a square preserves concavity. To check the two endpoint values, (18) also gives

$$7.32 < \alpha < 7.34, \qquad w_1 < 0.473, \qquad C_1 < 4.628, \qquad q_1 < 0.3878.$$

For $r = 1 - v$, the bracket is at least

$$K = r\left[\frac{46}{81}(1+v) - 0.473 r \right] - 7.34 \cdot 0.01022 v^2.$$

At both endpoints $K > 0$. Substitution gives the following exact values for $4(13.08)(7.32)K - (4.628r + 7.34)^2$.

v	lower bound for $F(v)$
9/25	11549877034957/2197265625000
13/20	5154805933931/468750000000

Both are positive, so concavity gives $F(v) > 0$ throughout. Since $F(v) = (1-v)^2[4kDa - (C_1 + D)^2]$, this gives $k > 0$ and $2\sqrt{kDa} > C_1 + D$. Applying the arithmetic–geometric mean inequality in (21) gives a contradiction.

Second range, $13/20 \le v \le 3/4$. Here we retain the positional term in (8). For the probability measure $d\mu/L$, write $X = \mathbb{E}x$, $N = \mathbb{E}\nu$, and $T = M/L$. Let H be a symmetric matrix of Frobenius norm one. Let H_0 be H minus the midpoint of its extreme eigenvalues times the identity. Then $T : H = T : H_0$ since T is trace free. The two extreme eigenvalues of H have difference at most $\sqrt{2}$, by Cauchy–Schwarz and $|H|_F = 1$. Consequently $\|H_0\|_{\mathrm{op}} \le 1/\sqrt{2}$. The pairing is

$$T : H_0 = X \cdot H_0 N + \mathbb{E}\big[(x - X) \cdot H_0(\nu - N)\big].$$

Cauchy–Schwarz bounds the second term using $\mathbb{E}|x-X|^2 = 1-|X|^2$ and $\mathbb{E}|\nu-N|^2 = 1-v^2$. Taking the supremum over H gives

$$|T|_F \le \frac{1}{\sqrt{2}}\left(|X|v + \sqrt{(1-|X|^2)(1-v^2)}\right).$$

We can use symmetric H because T is symmetric. The square of this bound minus $|X|^2$ is the quadratic form at the unit vector $(|X|, \sqrt{1-|X|^2})$ of the matrix $\frac{1}{2}(v, \sqrt{1-v^2})^{\otimes 2} - \operatorname{diag}(1,0)$. It is at most the larger eigenvalue, $(\sqrt{9-8v^2}-1)/4$. By (8), the mean of $x\cdot y$ with respect to the normalized defect pair measure is therefore at least

$$-\frac{\sqrt{9-8v^2}-1}{4(1-v^2)} = -\frac{2}{1+\sqrt{9-8v^2}} \ge -\frac{2}{3}.$$

Its unnormalized mass is $L^2(1-v^2)$. The tangent bound for B gives $P \ge \frac{2}{3}(1-q_1)L^2(1-v^2) > 0.408L^2(1-v^2)$. Then (20) gives $C_1 \ge L[f_0\,0.408(1+v) - 0.473(1-v)]$. The bracket is increasing and is positive at $v = 13/20$. Using $L \ge 4\pi$ would require

$$C_1 \ge 4\pi\left[f_0\,\frac{33}{20}\,0.408 - 0.473\,\frac{7}{20}\right].$$

Substitution of $\pi > 333/106$ bounds the right-hand side below by $4.628 + 52777/106000$, which contradicts $C_1 < 4.628$.

Third range, $3/4 \le v < 1$. The bound $B \ge 1/3$ gives $C_1 \ge L[f_0(1+v)/3 - 0.473(1-v)]$. This bracket is increasing and is positive at $v = 3/4$. Hence

$$C_1 \ge 4\pi\left[\frac{f_0}{3}\,\frac{7}{4} - \frac{0.473}{4}\right].$$

The same lower bound on π makes the right-hand side greater than $4.628 + 124307/954000$. This contradiction completes the proof. □

4.3. **Return to positive order.**

Proof of Theorem 1.2. If the theorem failed, there would be nonflat cones at orders $p_i \downarrow 0$. Proposition 2.2 gives a limit phase, and Proposition 4.2 makes it a hemisphere. Write e_* for its constant conormal. Weak convergence of the conormal measures gives

$$\mu_i(S^3) \longrightarrow 4\pi, \qquad \int |\nu_i - e_*|^2\, d\mu_i \longrightarrow 0.$$

The uniform quadratic cap bound also holds for radii larger than two by the bound on the total masses. Fix $\bar{p} \in (0,1)$ and discard the terms with $p_i > \bar{p}$. One constant $\Lambda > 1$ then gives the cap bound and the positive lower mass bound $\mu_i(S^3) \ge \Lambda^{-1}$ required in [15, Theorem 4.2]. The displayed conormal convergence gives its smallness hypothesis for large i. The theorem makes those cones halfspaces, which is a contradiction. □

5. From flat cones to rigidity

5.1. **The cone threshold.**

We use p for the parameter s of Caselli–Florit-Simon–Serra. In their Euclidean setting the kernel is $c_{d,p}|x-y|^{-d-p}$, where $c_{d,p}$ is the kernel coefficient of the Fourier multiplier $|\xi|^p$ in dimension d [8, Remark 2.4]. Their relative perimeter counts both orders of each opposite-phase pair. Thus

$$\operatorname{Per}_p^{\mathrm{CFS}}(E;D) = 2c_{d,p}\operatorname{Per}_p(E;D). \tag{22}$$

The factor follows directly from [8, Definition 1.3] and is unchanged by deformation.

Lemma 5.1 (The dimension threshold)**.** *Fix $0 < p < 1$. If the nontrivial stationary stable p-perimeter cones with smooth compact links and locally one-sided boundary off zero are halfspaces in dimensions three and four, then $n_p^* > 4$ in the sense of [8, Definition 1.11].*

Proof. Consider a nonflat smooth stable cone in the CFS convention, in dimension $d = 3$ or 4. We use the density-one representative and its phase interface, following the essential-boundary convention in [8, proof of Lemma 4.6]. A null or conull phase has no such interface, so both phases have positive measure. In a smooth boundary chart off zero, each connected open side of the sheet lies in a single phase. If the two sides have the same phase, the sheet is null and every point in the chart has the density of that phase. If the sides have opposite phases, points on the sheet have density $1/2$, by rescaling the graph to its tangent plane. Thus the density-one representative retains the open phase side at each such sheet. Dilation preserves density, so the representative is a cone. The volume fraction in every ball centered at the vertex is the fraction in B_1, which lies strictly between zero and one. The representative therefore excludes the vertex and is open. Since both phases occur and $\mathbb{R}^d \setminus \{0\}$ is connected, there is a regular boundary point. Its dilation orbit meets S^{d-1}. At every such intersection the radial vector is tangent to the boundary and normal to the sphere. This makes the intersection transverse, so the link is smooth. It is compact because it is closed in S^{d-1}.

Smooth diffeomorphisms preserve null sets, so taking this representative leaves the variations unchanged. For any compact smooth field away from zero, choose a bounded annulus containing its support and apply (22) along the autonomous flow of that field. Stationarity and stability in [8, Definitions 1.6–1.7] give the corresponding first and second variation conditions for our perimeter. The first variation depends only on the initial velocity. At stationarity, Lemma 2.1 gives the same property for the second variation. Thus the cone meets the conditions for smooth deformations supported away from zero. It agrees almost everywhere with a halfspace by hypothesis, and its density-one representative is that open halfspace. The minimum in [8, Definition 1.11] runs over $d \geq 3$. Excluding three and four gives $n_p^* \geq 5$. □

Take $p_3 > 0$ from Tsiamis's dimension-three cone theorem [15, Theorem 1.1], using the normal stability convention in Lemma 2.1. Take p_0 from Theorem 1.2 and set

$$s_* = \tfrac{1}{2} \min\{p_3, p_0, 1/2\}. \tag{23}$$

Then Lemma 5.1 gives $n_p^* > 4$ for all $0 < p < 2s_*$.

In the proof of [8, Theorem 4.12], a blow-up at a nonvertex singular point of a cone in $\mathbb{R}^4$ splits off a line. The density estimates in their Lemma 4.13, applied to both phases at the blow-up center, keep the limiting cone nontrivial. If the base of this cylinder is smooth away from its own vertex, the possible singularities of the cylinder lie on its axis. We use the following observation in that case.

Lemma 5.2 (Variations of a cylinder and its base)**.** *Let $0 < p < 1$ and let $F \subset \mathbb{R}^3$ be a nontrivial cone with a smooth compact link and a locally one-sided boundary off zero. If $F \times \mathbb{R}$ is stationary and stable for the p-perimeter away from the ambient origin, then F is stationary and stable away from its vertex.*

Proof. The base has locally finite classical perimeter by Lemma 2.1. The product does as well, since its perimeter measure on bounded cylinders is the product of the base perimeter measure and length measure, by Fubini in the distributional derivative. The translation estimate in that lemma then gives finite localized p-perimeter. Fix a smooth compactly supported field X on $\mathbb{R}^3 \setminus \{0\}$, with flow ϕ_τ, and let η be a nonnegative, nonzero smooth function compactly supported on $\mathbb{R}$. For $L \geq 1$ use the field $(\eta(t/L)X(x), 0)$ at $(x, t) \in \mathbb{R}^3 \times \mathbb{R}$. Its support misses the ambient origin. Its flow takes (x, t) to $(\phi_{\tau\eta(t/L)}(x), t)$. In the pullback of the perimeter, the Jacobian is the determinant of the horizontal derivative of this map.

For the first or second variation, divide by L and change variables from the two vertical coordinates t, t' to $w = t/L$ and $h = t' - t$. For fixed x, y, w, h, the two vertical cutoff values $\eta(w)$ and $\eta(w + h/L)$ converge to the same value. The derivative of order $j = 1, 2$ of the pair integrand therefore converges to $\eta(w)^j$ times the derivative with the same horizontal flow at both endpoints. To justify the limit, the pullback estimates in Lemma 2.1 apply with uniform constants because the lifted fields have uniform bounds on their derivatives for $L \geq 1$. The derivative vanishes unless one horizontal point lies in $\operatorname{supp} X$ and at least one of $w, w+h/L$ lies in a fixed compact interval containing $\operatorname{supp} \eta$. For fixed h this restricts w to a bounded interval independent of L, and the length of the union of the two allowed intervals is uniformly bounded. After integration in w, the absolute derivative is thus bounded by a constant times

$$(|x - y|^2 + h^2)^{-(4+p)/2}$$

on opposite-phase base pairs with at least one endpoint in $\operatorname{supp} X$. Integrating in h gives a constant times $|x - y|^{-3-p}$. This is integrable there by finite localized perimeter of F from Lemma 2.1. The same bounds apply before division by L for each fixed L, so the perimeter derivatives can be taken under the integral. Dominated convergence, first in w and then in x, y, h, now gives the limit.

For the limiting derivative, integration in h gives the base kernel times the positive constant $\int_{\mathbb{R}} (1 + h^2)^{-(4+p)/2}\, dh$. Integration in w gives $\int_{\mathbb{R}} \eta(w)^j\, dw > 0$. Thus the divided derivative of order j converges to a positive multiple of the corresponding base derivative. The first derivative of the cylinder vanishes and the second is nonnegative. This proves stationarity and stability of the base, first for autonomous flows and then for smooth deformations by Lemma 2.1. □

In [8, proof of Theorem 4.12], the cone with one invariant direction is stationary and is stable away from its vertex by their Lemma 4.15. If its base is smooth off zero, use its density-one representative as in Lemma 5.1. Lemma 5.2 then gives a stable three-dimensional cone. Since $3 < n_p^*$, this base is flat. The cylinder is therefore a halfspace, and improvement of flatness [8, Theorem 4.14] contradicts singularity at the blow-up center. In the other case there is a nonzero singular point of the base. Blowing up at the corresponding point on the zero-height slice preserves the vertical invariance and adds the radial direction at that point, as in [8, proof of Theorem 4.12]. The two directions are independent, so the resulting cone is flat by property (6) of [8, Theorem 4.12], again contradicting singularity at the blow-up center. This covers the dimension reduction used in both finite Morse index theorems in dimension four.

5.2. **The Allen–Cahn equation and its index.** Put $F(t) = t - t^3$ and $\mathcal{W}(t) = (1 - t^2)^2/4$. For a bounded open set D in dimension d, the Fourier-normalized energy is

$$\mathcal{E}_d(u; D) = \frac{c_{d,p}}{4} \iint_{\mathcal{Q}_D} \frac{|u(x) - u(y)|^2}{|x - y|^{d+p}}\, dx\, dy + \int_D \mathcal{W}(u)\, dx, \qquad \mathcal{Q}_D = (\mathbb{R}^d)^2 \setminus (D^c)^2. \tag{24}$$

A minimizer on D minimizes this energy among functions with the same values outside D. On a real compact smooth test ϕ, twice differentiating the square gives kinetic coefficient $c_{d,p}/2$ on the ordered pairs. By Plancherel the second variation is

$$Q_u(\phi) = \int_{\mathbb{R}^d} |\xi|^p |\widehat{\phi}(\xi)|^2\, d\xi + \int_{\mathbb{R}^d} (3u^2 - 1)\phi^2\, dx. \tag{25}$$

Here we use the unitary Fourier transform. The Morse index is the supremum of the dimensions of real subspaces of test functions on which Q_u is negative definite. Any finite basis is supported in one ball, so the global index equals the supremum of the indices on balls. In any test domain the index is unchanged if we use compactly supported C^1 or Lipschitz tests. Indeed, mollify the members of a finite basis of such tests, with mollifier radii small enough to keep the supports in a common compact subset of the domain. The approximating functions have a common Lipschitz bound. In the pair form of the kinetic term, at least one

point belongs to that compact set. Products of two test differences are $O(|x-y|^2)$ near the diagonal and bounded at large distances. Dominated convergence for the polarized form gives convergence of all matrix entries. A negative definite matrix is strictly negative on the unit sphere of coefficient vectors. This remains true for sufficiently close matrices, and implies that the approximating tests are linearly independent.

Lemma 5.3 (Bounds and increasing profiles)**.** *Let $d \geq 2$ and $0 < p < 1$. A bounded entire classical solution of $(-\Delta)^{p/2}u = F(u)$ in $\mathbb{R}^d$ is a constant well or has values in $(-1,1)$. It has uniform $C^{2,\alpha}$ bounds on balls of fixed radius, and limits of its translates are solutions that inherit stability. The constant zero has infinite Morse index. If a directional derivative of u is nonnegative and is positive somewhere, it is positive everywhere and u is stable. If u depends on one coordinate through a nonconstant increasing profile, a translate of that profile satisfies the one-dimensional equation and normalization in Theorem 1.1.*

Proof. The interior estimates in [5, Proposition C.1], applied on unit balls, give uniform $C^{2,\alpha}$ bounds there, with constants independent of the center. In particular, the first two derivatives are bounded on $\mathbb{R}^d$. Any sequence of translates has a subsequence converging in C^2_{loc}. The limit solves the same equation. Indeed, for the kernel integral written with difference variable $z = y - x$, the integrand is bounded by $C|z|^{1-d-p}$ for $|z| < 1$ and by $C|z|^{-d-p}$ for $|z| \geq 1$. If u is stable, so is the limit, by convergence on the support of each fixed test in (25).

If $\sup u > 1$, translate points where u approaches its supremum to the origin. A subsequence gives a solution with maximum $M = \sup u$ at zero. The kernel integral at the maximum is nonnegative, while $M - M^3 < 0$. Applying this argument to $-u$ shows that $-1 \leq u \leq 1$. If u reaches either well, the kernel integral there has one sign and is zero. Since the kernel is strictly positive, u must be a constant well.

Let $w = \partial_e u \geq 0$ be a directional derivative which is positive somewhere. It satisfies $(-\Delta)^{p/2}w = F'(u)w$. Differentiation under the kernel integral uses the same bounds as above, now for the difference of the first derivatives, since D^2u is bounded. If w vanished at some point, the kernel integral there would be strictly negative, whereas the right-hand side would be zero. Thus $w > 0$. Multiply this equation by ϕ^2/w, a compactly supported C^1 function since $w > 0$. Integrate and symmetrize the kernel term. For positive w_x, w_y one has

$$(\phi_x - \phi_y)^2 - (w_x - w_y)\left(\frac{\phi_x^2}{w_x} - \frac{\phi_y^2}{w_y}\right) = w_x w_y \left(\frac{\phi_x}{w_x} - \frac{\phi_y}{w_y}\right)^2.$$

Integrating this identity against $(c_{d,p}/2)|x-y|^{-d-p}$ then proves that (25) is nonnegative. All symmetrized integrals converge absolutely by the bounds on w, its first derivatives, and the compactly supported test.

The only constant solutions are the wells and zero. The wells are stable. Zero has infinite Morse index: for any finite-dimensional subspace of compact smooth tests, replace every $\phi(x)$ by $\phi(x/T)$. After division by T^d, its form at zero is

$$T^{-p}\int_{\mathbb{R}^d} |\xi|^p |\widehat{\phi}(\xi)|^2 \, d\xi - \|\phi\|_2^2.$$

The kinetic term is bounded on the L^2 unit sphere of the original subspace. For large T the dilated subspace is negative, and its dimension can be arbitrarily large.

For use here and below, let $1 \leq k < d$. A solution independent of $d-k$ coordinates solves the equation on $\mathbb{R}^k$ with the same Fourier normalization. Indeed, writing $h \in \mathbb{R}^k \setminus \{0\}$ for the retained difference and integrating in the other coordinates gives

$$c_{d,p}\int_{\mathbb{R}^{d-k}} (|h|^2 + |z|^2)^{-(d+p)/2} \, dz = c_{k,p}|h|^{-k-p}.$$

Scaling gives the power of $|h|$, and the coefficient follows from the symbol formula at $(\xi', 0)$ by Tonelli applied to $1 - \cos(h \cdot \xi')$. The original kernel integral converges absolutely for bounded Lipschitz functions since $p < 1$.

Now suppose $u(x) = Q(e \cdot x)$ with a nonconstant nondecreasing profile Q. Its directional derivative is nonnegative and positive somewhere, so $Q' > 0$ and u is stable. The two distinct limits of Q at infinity arise as translates of u. They are stable constant solutions, hence are -1 and 1 in that order. The kernel reduction with $k = 1$ gives the one-dimensional equation. At the unique zero a of Q, set $q(t) = Q(t + a)$. This gives the normalized layer. □

5.3. **Finite Morse index in dimension four.**

Lemma 5.4 (Flat slabs at large radii)**.** *Let $d \geq 2$, $0 < p < 1$, and let $u : \mathbb{R}^d \to (-1, 1)$ be an entire solution of $(-\Delta)^{p/2}u = u - u^3$. Suppose that every sequence of radii tending to infinity has a subsequence along which the inclusions*

$$\{e \cdot x \leq -aR\} \subset \{u \leq -4/5\} \subset \{u \leq 4/5\} \subset \{e \cdot x \leq aR\} \quad \text{in } B_R \tag{26}$$

hold eventually for each fixed relative width $a \in (0, 1)$ with some unit directions e. Then u is one-dimensional with an increasing profile.

Proof. Let $b(R)$ be the infimum of the admissible widths $a \in [0, 1]$ in (26), allowing the direction to depend on R. Width one is admissible in the open ball. If $b(R)$ did not tend to zero, some sequence tending to infinity would have $b(R)$ bounded below by a positive constant. Its subsequence in the hypothesis contradicts this bound. For large R, take $a(R) = \sup_{\rho \geq R} b(\rho) + R^{-1}$, and take $a(R) = 1$ below a radius at which the large-radius formula is at most one. Widening the slab preserves the inclusions, so $a(R) > b(R)$ is admissible wherever we use the formula. This gives a positive nonincreasing width tending to zero at all radii, as required in [8, Theorem 4.22]. □

Lemma 5.5 (Classification below the cone threshold)**.** *Let $0 < p < 1$ and $4 < n_p^*$. Every bounded entire classical solution of $(-\Delta)^{p/2}u = u - u^3$ in $\mathbb{R}^4$ with finite Morse index is a constant well or a normalized layer up to translation. All these solutions are stable.*

Proof. By Lemma 5.3, a constant solution of finite index is a well, and a nonconstant one has values in $(-1, 1)$. In Euclidean space the energy in [8, Definition 1.13 and Remark 2.4], with their order equal to p and $\varepsilon = 1$, is exactly (24). The function u has finite energy on bounded domains by its boundedness and Lipschitz bound. The first variation along a compact test ϕ equals $\int \phi[(-\Delta)^{p/2}u + \mathcal{W}'(u)]$, by symmetrizing the kernel integral. Thus u is a critical point on every bounded domain. The index in [8, Definition 1.15] is that of the Hessian of this energy, which is (25).[1] The compact C^1 tests give the same index by the matrix approximation above. The global bound therefore gives the uniform local index bound used in the Euclidean proof of [8, Theorem 1.32].

We give the limiting step of that proof at all large radii. After a translation we can arrange $|u(0)| < 9/10$. Otherwise continuity on the connected space would give $u \geq 9/10$ everywhere or $u \leq -9/10$ everywhere. In the first case apply the maximum-translate argument of Lemma 5.3 to $-u$, and in the second to u. The resulting maximum M is in $(-1, 0)$, where $M - M^3 < 0$, a contradiction.

Take any sequence $R_i \to \infty$ and write $u_i(x) = u(R_i x)$. It solves the CFS equation with $\varepsilon_i = 1/R_i$. The energy with potential $\varepsilon_i^{-p}\mathcal{W}$ on a domain D is R_i^{p-4} times the original energy on $R_i D$, under pullback by dilation. This identifies the local second variations and preserves the global index. The Euclidean local compactness in [8, Theorem 1.30 and Section 4.6] gives, by a diagonal subsequence, a limit $u_i \to 2\mathbf{1}_E - 1$ in L^1_{loc}. The flat metric hypothesis

[1] In arXiv version 3 of [8], differentiating the energy (4) twice gives $1/2$ in front of the ordered pair integral with their kernel. The display (6) retains the energy coefficient $1/4$. The mixed term for disjoint tests in the proof of [8, Lemma 3.10] uses the Hessian coefficient $1/2$. In the potential estimate in [8, Lemma 3.12], the corresponding bound uses $2\mathcal{E}^{\mathrm{Sob}}$, which changes only the constant in that estimate.

of [8, Proposition 1.29] holds in Euclidean balls, and the constants are uniform in the center. If the limit were constant on a ball centered at zero, the density estimate on that ball would force $u_i(0) > 9/10$ or $u_i(0) < -9/10$ for large i. Hence E is nontrivial and zero lies on the interface. The limit belongs to the Euclidean class $\mathcal{A}_m(\mathbb{R}^4)$ of [8, Section 4.6], where $m \geq 1$ is a fixed index bound. Their Theorem 4.12 applies to that class in dimension four, with the cylinder step above, and gives a halfspace with boundary through zero.

Let e be its inward unit normal. For fixed $\delta \in (0,1)$, each point of B_1 with $|e \cdot x| \geq \delta$ has a ball of radius $\delta/2$ inside B_2 on the same side of the plane. On such balls the L^1 difference from the corresponding well tends uniformly to zero, bounded by the full $L^1(B_2)$ error. By [8, Proposition 1.29], u_i is below $-9/10$ on the negative cap and above $9/10$ on the positive cap for large i. Thus the subsequence admits arbitrarily small relative widths in (26). The sequence of radii was arbitrary, so Lemma 5.4 gives the increasing profile in Theorem 1.32 of [8]. Lemma 5.3 gives its normalization and stability. □

Proof of Corollary 1.3. If $0 < s < s_*$, then $4 < n^*_{2s}$ by (23) and Lemma 5.1. Apply Lemma 5.5 at $p = 2s$. □

Proof of Theorem 1.4. By Lemma 2.1, E has finite localized perimeter. For a compact smooth field X, let $Q_E(X)$ be the second derivative of the localized perimeter along its autonomous flow. This quadratic form is independent of the localization containing the support. By (22), the first variation in the CFS convention vanishes and its second variation is $2c_{4,p}Q_E(X)$. This also holds with support in the closure of a bounded Lipschitz domain, by Lemma 2.1.

Let m be the index of Q_E. Given $m+1$ test fields in any such domain, a linear dependence gives a nonzero coefficient vector with zero combined field. If they are independent, their span has dimension $m+1$ and contains a nonzero field with nonnegative Q_E. In both cases, rescale the coefficient vector to have length one. Conversely, a negative subspace of dimension $m+1$ would have a basis supported in one ball and would violate this condition in that ball. Thus m agrees with the geometric index in [8, Definitions 1.6–1.7]. These are the stationarity and weak index conventions used in their Euclidean class in [8, Definition 4.23 and Proposition 4.24]. The constant sequence $E_j = E$ belongs to that class.

We have $4 < n^*_p$ by (23) and Lemma 5.1. Both phases have positive measure in a chart about any boundary point, so the nonempty C^2 interface satisfies the hypotheses of [8, Theorem 1.31]. The dimension reduction in that theorem uses the cylinder case proved above. It yields a halfspace. Every boundary point of the original phase has density $1/2$ in its one-sided graph chart. Interior points have density one and exterior points have density zero. Thus E is the density-one representative, and its agreement with the halfspace holds as open sets. □

5.4. **Monotone solutions in dimension five.** We now use the stable case of Lemma 5.5 on the two vertical limits. The reduction works at any fixed order with the stated cone hypothesis.

Proposition 5.6. *Fix $p \in (0,1)$. Suppose every nontrivial stationary stable p-perimeter cone in $\mathbb{R}^3$ or $\mathbb{R}^4$ with smooth compact link and locally one-sided boundary away from the vertex is a halfspace, up to null sets. Then every bounded entire classical solution of*

$$(-\Delta)^{p/2}U = U - U^3 \quad \text{in } \mathbb{R}^5, \qquad \partial_{x_5}U > 0,$$

is a layer of the form in Theorem 1.1, with $s = p/2$.

Proof. The vertical limits. By Lemma 5.3, we have $|U| < 1$ and U is stable. Write $x = (x', x_5)$ and let $U^{\pm}(x') = \lim_{t\to\pm\infty} U(x', t)$, which exist by monotonicity. The local estimates give convergence of the vertical translates in C^2_{loc} to these limits viewed as functions on $\mathbb{R}^5$. Indeed, the pointwise limit fixes every subsequential limit. Thus the lifted limits solve the equation

and are stable on $\mathbb{R}^5$. They are independent of x_5, so the kernel reduction in Lemma 5.3 gives the equation on $\mathbb{R}^4$.

To see stability in dimension four, take $\phi \in C_c^\infty(\mathbb{R}^4)$ and $\psi \in C_c^\infty(\mathbb{R})$ with $\|\psi\|_2 = 1$. Use $\phi(x')T^{-1/2}\psi(x_5/T)$ in the five-dimensional stability inequality for the lifted limit. The kinetic multiplier in the resulting integral is $(|\xi'|^2 + T^{-2}\tau^2)^{p/2}$, which tends to $|\xi'|^p$. For $T \geq 1$ it is bounded by $|\xi'|^p + |\tau|^p$. The potential term is independent of T, so dominated convergence gives stability on $\mathbb{R}^4$. The cone hypothesis implies $4 < n_p^*$ by Lemma 5.1. Lemma 5.5 then shows that each $U^\pm$ is a well or an increasing layer.

Minimization. We apply Cabré–Cinti–Serra, whose kernel coefficient is $2-p$ [5, Definition 1.12]. Set $u(X) = U(\lambda_5 X)$ with $\lambda_5^p = c_{5,p}/(2-p)$. The operator scales by λ_5^p, so u solves their equation. In their energy on D, the pair change of variables gives a factor λ_5^{p-5} and the potential term a factor λ_5^{-5}. Since $(2-p)\lambda_5^p = c_{5,p}$, their energy of u on D is $\lambda_5^{-5}\mathcal{E}_5(U;\lambda_5 D)$. This identifies stability and minimization in the two normalizations. The two vertical limits of u are $U^\pm(\lambda_5 X')$. Their lifts solve the equation in the normalization of u, since they are limits of its vertical translates. They are wells or increasing one-dimensional solutions. By [5, Proposition 7.9], u is an entire minimizer. Scaling back, U minimizes (24) in every bounded domain.

Flat limiting interfaces. We first check that the cone hypothesis also gives flatness for all nontrivial minimizing cones in $\mathbb{R}^3$ and $\mathbb{R}^4$. At a regular point of a minimizing cone the boundary is a $C^{1,\alpha}$ graph [7, Section 6]. Centering coordinates there, rotating the tangent plane to horizontal, and shrinking the ball make the relative height of the graph as small as needed in [1, Corollary 2], with lower order bound $p/2$. That result makes the graph smooth. In the representative of [7, Section 4] each boundary ball contains both phases in positive measure. Thus in a graph chart the phases occupy opposite sides. A minimizing cone regular off zero therefore has a smooth compact link and is stationary and stable in the sense of Lemma 2.1.

For a minimizing cone in $\mathbb{R}^3$, the singular set has Hausdorff dimension at most zero by [13, Corollary 2]. A nonzero singular point would give a whole singular ray by conicality. Thus the cone is regular away from the origin and the cone hypothesis makes it a halfspace. In $\mathbb{R}^4$, suppose there is a nonzero singular point. Dilate and rotate to put it at the fourth coordinate unit vector. By [7, Theorem 10.3], a blow-up there splits off a line and has a three-dimensional minimizing cone as base, singular at its vertex. This contradicts the classification just proved. Thus the four-dimensional cone is also regular away from zero, and the cone hypothesis makes it a halfspace.

Let $R_i \to \infty$. Pass to an increasing subsequence and apply [5, Proposition 6.1 and Theorem 2.10] to u at the radii R_i/λ_5. Along a subsequence,

$$U(R_i x) \longrightarrow 2\mathbf{1}_C(x) - 1 \quad \text{in } L^1_{\mathrm{loc}}(\mathbb{R}^5) \tag{27}$$

for a nontrivial cone C. On the same subsequence we retain the geometric convergence of the superlevel sets in [5, equation (2.10)].

To prove that C minimizes the p-perimeter, use the exponent $p/2 < 1/2$ of [14]. Their unit-scale energy has kinetic coefficient $1/2$ on the ordered pairs of (24). Taking their potential to be $2\mathcal{W}/c_{5,p}$ makes their energy $2/c_{5,p}$ times ours. It has the required two wells. With $\varepsilon_i = 1/R_i$, the scaled energy of $U(R_i\,\cdot)$ on B_ρ equals the unit-scale energy of U on $B_{\rho R_i}$ times $R_i^{-(5-p)}$. Competitors with the same exterior values correspond under dilation. Hence the rescaled functions minimize their scaled energy, and [14, Theorem 1.3] bounds this energy on B_ρ for large i, using minimality on $B_{\rho R_i + 2}$. By (27) and the uniform bound by one, the rescaled functions converge weakly-* in $L^\infty(B_\rho^c)$ to $2\mathbf{1}_C - 1$. To see this, first test against bounded functions of compact support and then approximate in L^1. Thus [14, Theorem 1.2(i)] applies on each ball along this subsequence, with the indicated exterior values. On a function of the form $2\mathbf{1}_C - 1$, the limit energy is $4\,\mathrm{Per}_p(C;B_\rho)$ since the squared difference is four on the opposite-phase pairs and the factor $1/2$ counts each pair once. It follows that

C minimizes on every ball. This covers every bounded change of its phase by taking a ball containing the change.

Write $\mathbf{e}_5 = (0,0,0,0,1)$. For each fixed $t > 0$, monotonicity and local L^1 convergence give $\mathbf{1}_C(x) \le \mathbf{1}_C(x + t\mathbf{e}_5)$ for almost every x. Integrating this inequality on balls centered at x shows that density one at x implies density one at $x + t\mathbf{e}_5$. Thus $C + t\mathbf{e}_5 \subset C$ holds exactly for the density-one representative, for each $t > 0$. Positive dilations preserve density, so this representative is also a cone. By [5, Theorem 2.10, equation (2.9)], a boundary point of the representative used there has densities bounded away from zero and one. A point of density one must therefore be interior, and a point of density zero exterior. Thus taking the density-one representative gives the open phase, with both phases of positive measure in every ball about a boundary point. This meets the phase convention in [10, Section 2]. By [10, Proposition 3.1], C is a halfspace or is a cylinder in the fifth direction. In the cylinder case, invariance under all vertical translations gives $C = D \times \mathbb{R}$ with $D = \{x' \in \mathbb{R}^4 : (x', 0) \in C\}$. The factor is an open cone by openness and dilation invariance of C. It minimizes the p-perimeter by [7, Theorem 10.1]. Both phases of D have positive measure, since both phases of C do. Hence D is a halfspace by the minimizing cone classification above. In either case C is a halfspace with boundary through zero, since C is a cone. Figure 3 shows the two alternatives.

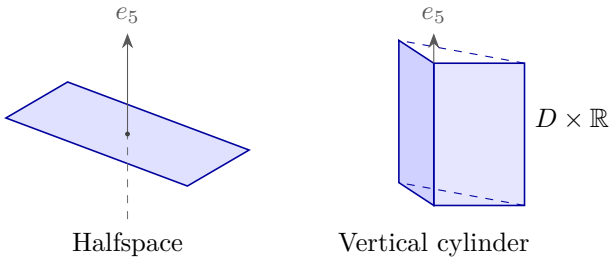


FIGURE 3. Two possible shapes for a minimizing blow-down with an ordered vertical direction. The halfspace boundary is drawn in a finite window. The cylinder repeats the same base cone D at every height. In the proof $C \subset \mathbb{R}^5$ and $D \subset \mathbb{R}^4$. The drawing uses lower-dimensional models.

From flat limits to a layer. We have shown that every sequence $R_i \to \infty$ has a subsequence with a halfspace limit. Fix such a subsequence and let e be the inward unit normal of the limiting halfspace. Use the geometric convergence from the extraction in (27) for u. Apply [5, Theorem 2.10, equation (2.10)] to $-u$ along the same radii and take a further subsequence. The solution $-u$ is stable because the cubic is odd and its derivative is even. By uniqueness of the local L^1 limit, its limiting phase is the complementary halfspace. For both signs, use the threshold $-4/5$ in B_2. Points in B_1 at distance at least $\delta > 0$ from the corresponding halfspace cannot belong to the superlevel set for large i. Thus

$$\begin{aligned} U(R_i x) &< -4/5 \quad \text{if } x \in B_1 \text{ and } e \cdot x \le -\delta, \\ U(R_i x) &> 4/5 \quad \text{if } x \in B_1 \text{ and } e \cdot x \ge \delta. \end{aligned}$$

The choice of phase representative on the limiting plane does not affect these distance estimates.

After rescaling to B_{R_i}, these are the slab inclusions (26) for U. Since the initial sequence was arbitrary, Lemma 5.4 gives one-dimensionality.

Write $U(x) = Q(e \cdot x)$. The direction has $e_5 \neq 0$, since otherwise U would be independent of x_5. Choose its sign so that $e_5 > 0$. Then $\partial_{x_5} U = e_5 Q' > 0$ makes Q increasing. Lemma 5.3 gives the normalized layer and shift of Theorem 1.1. □

Proof of Theorem 1.1. For $0 < s < s_*$ with s_* as in (23), the cone classifications of Tsiamis [15, Theorem 1.1] and Theorem 1.2 give the hypothesis of Proposition 5.6 at $p = 2s$. □

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